Koa Shen

Mechanical Engineer
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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Dynamics, Circuits, Mechanics of Materials, MATLAB. Current: Dynamical Systems and Machine Learning. Planned: Mechatronics, Robotics Lab, Inverse Kinematics.

EXPERIENCE

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Built an open-source Python/Drake digital twin for a 30-DOF, 305-body beamline motion assembly of linear, rotary, and tip-tilt stages; millisecond collision checks exposed at least five physical pinch points.
- Integrated OpenCascade CAD import, CoACD convex decomposition, and MeshCat visualization so engineers can load and inspect an assembly in about 30 seconds.
- Created 15 reusable motion-stage models so new assemblies can be simulated without rewriting kinematics; now being adopted across three LCLS assemblies, with path planning and EPICS integration in development.
- Redesigned an optics assembly with three kinematic bases, six hot-swappable optics, and two added DOF for detector maneuverability; checked stage and mount load cases for $100k+ detectors.

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation; the car passed FSAE technical inspection.
- CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding bearing fits within 0.0005 in.
- Helped develop a steering wheel integrating buttons, potentiometers, a display, and a quick release; FEA-guided design reduced mass from 5 lb to 1.7 lb.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Maintained electromechanical exhibits, including an encoder-controlled spinning table and a piston-crank drive; led three major overhauls across 100+ serviced exhibits.
- Redesigned rotary-encoder mounts to resist garnet-sand ingress and simplify maintenance; manually machined replacements onsite.

PROJECTS

Closed-Loop Cycloidal Actuator | Personal Project — Summer 2026 - Present

- Designed a 15:1 cycloidal reducer that fits within a NEMA 17 motor's 42 mm square face and is shorter axially than the motor, so the joint packages into linkages without growing its envelope.
- Wrote PlatformIO C++ firmware for a TMC2209 driver, including phase-based bring-up, step/dir generation, and real-time CSV telemetry.
- Closed the loop with a 12-bit AS5600 encoder on an Arduino Uno; bench-validated automatic position-error correction and missed-step detection.
- Built as the first joint of a planned 6-DOF arm; Drake kinematic simulation and trajectory optimization are next.

FSAE Chassis Welding Fixtures | Gaucho Racing — Winter 2024 - Spring 2026

- Designed fixtures that located 81 chassis tubes within 0.050 in of CAD nodes; cut assembly time 50% for GR25 and a further 30% for GR26.

TECHNICAL SKILLS

Robotics and Controls: Drake, kinematics, collision detection, closed-loop position control, stepper motors, TMC2209, AS5600 encoders, I2C
Software: Python, C/C++, PlatformIO, Arduino, MATLAB, MeshCat, OpenCascade, Git
Mechanical: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, FEA, CNC/manual machining, 3D printing